

Ali Mansouri

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Professional Experience

Stratobotic

September 2024 - December 2024

On-Board Electronics Engineer (Project-Based Role)

Italy

- Gained 4 months of experience designing and optimizing PCB schematics and layouts for UAV power distribution using KiCAD.
 - Enhanced energy efficiency with Maximum Power Point Tracking (MPPT) circuits.
 - Programmed Arduino-based GNSS tracking systems for precise positioning.
 - Received a strong recommendation letter for outstanding performance during the internship.
- Tags:** PCB Design, UAV Electronics, GNSS, Power Optimization, KiCAD

P2CAM

August 2024 - April 2026

Embedded IoT Engineer (Part-Time Role)

Italy

- Led the end-to-end firmware architecture of a battery-powered IoT camera product, integrating 15+ sub-systems — from BLE wireless connectivity and animated display UI to intelligent power management — delivering a market-ready embedded platform on ESP32-S3.
 - Extended device field battery life to over 6 months by designing a multi-tier power management system, reducing idle power consumption by 99% and enabling always-ready operation for consumer use cases.
 - Engineered a custom BLE-to-mobile communication protocol with sub-100ms response time, enabling seamless smartphone pairing, remote camera control, and real-time device status synchronization.
 - Delivered production-grade firmware with OTA update capability, a multi-layered watchdog system, and optimized memory management across dual-core FreeRTOS — ensuring sustained 30 FPS animated UI and field reliability on resource-constrained hardware.
- Tags:** IoT Product Development, Embedded Systems, BLE Protocol Design, Power Management, LVGL, FreeRTOS, ESP32-S3

Politecnico di Torino Robotic Team

November 2023 - September 2024

Embedded Software Developer

Italy

- Accumulated 11 months of experience developing robotic functionalities using C and Python under competition constraints.
 - Implemented real-time operating systems (RTOS) on STM32 microcontrollers to optimize task management.
 - Configured communication protocols (USART, CAN, SPI) to enable robust data exchange between components.
- Tags:** Embedded Systems, Robotics, RTOS, STM32, Communication Protocols

Education

Master in Computer Engineering – Embedded & Smart System Design

Polytechnic University of Turin, 2025 - 2027

Key Courses: Computer Architectures, Electronics for Embedded Systems, Embedded Operating Systems and Architectures, Specification and Simulation of Digital Systems, Synthesis and Optimization of Digital Systems, Microelectronic Systems, Cybersecurity for Embedded Systems, System-on-Chip Architecture, Software Engineering

Bachelor in Electronics and Communication Engineering

Polytechnic University of Turin, 2022 - 2025

Key Courses: Applied Electronics, Circuit Theory and Applications, Automatic Control, Electronic Circuits, Electronic Devices, Communication Networks, Algorithms and Programming

Technical Proficiency

Areas of Expertise	Embedded Systems, IoT Solutions, Real-time Operating Systems, Robotics, PCB Design, BLE Communication, Debugging , Object-detection
Programming Languages	Proficient in Python, C, C++
Tools and Frameworks	FreeRTOS, Git, Docker, ESP-IDF,,Arduino , STMCubeIDE, STMCubeMX, Keil, Altium Design

Portfolio of most relevant projects

PCB Design	Designed and optimized UAV power distribution systems with MPPT for Stratobotic.
IoT Device Development	BLE-enabled IoT system with mobile app integration for P2CAM.
Robotics Development	Developed robotic functionalities and real-time systems for Politecnico di Torino Robotic Team.

Certificates

Microcontroller Embedded C Programming 2024	<i>Udemy</i>
Mastering RTOS: Hands-on FreeRTOS and STM32Fx 2024	<i>Udemy</i>
Algorithm and Data structure on C 2023	<i>Politecnico di Torino</i>